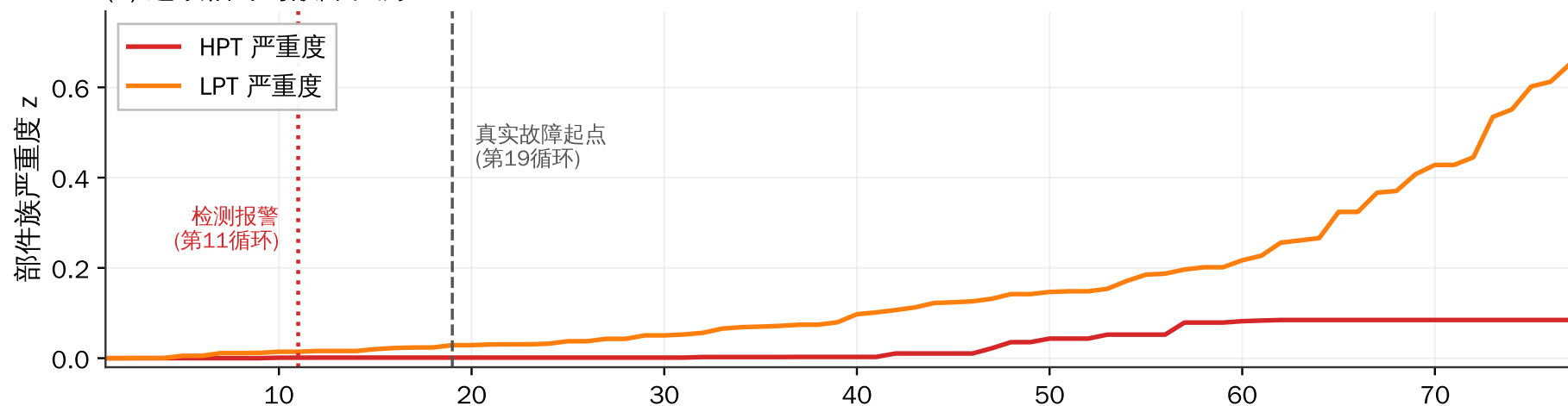
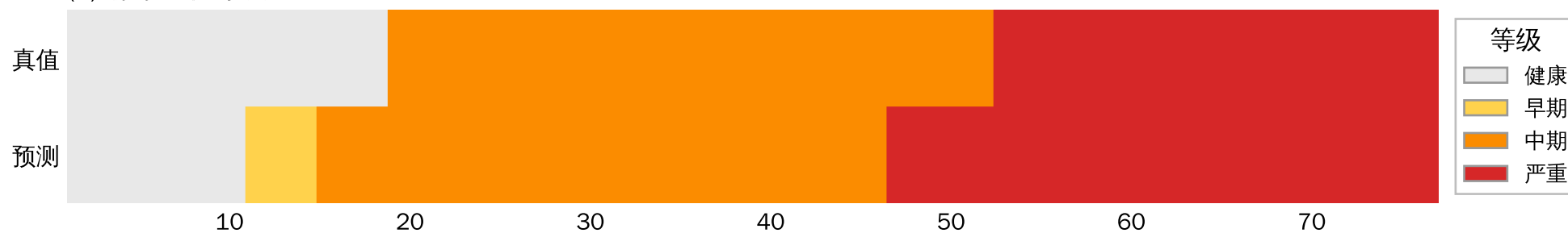


诊断闭环时间线：DS03 发动机 13 (未见故障组合，HPT+LPT 退化)

(a) 连续估计与故障检测



(b) 有序退化等级



(c) 部件族隔离

